Chun-Yi Lee

Data Scientist

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SUMMARY

Data scientist with a quantitative research background in statistical modelling, time-series analysis, and investment research. WorldQuant BRAIN Research Consultant (Gold level) with 38 submitted equity signals. Builds multi-agent research systems, RAG knowledge-base tools, and large-scale data pipelines that turn complex data into decision-ready reports, with stakeholder and budget experience from co-leading a US$37K nonprofit sponsorship campaign.

EXPERIENCE

Research Consultant (Conditional) | WorldQuant BRAIN Remote | Mar 2026 – Present

- Develop predictive signals for global equity markets under a paid consulting contract with WorldQuant, a global quantitative investment firm; reached Gold level in the BRAIN Genius Program
- Delivered 38 signals passing WorldQuant's quality and robustness checks (median in-sample Sharpe 1.63) across U.S., global, and European markets
- Built an automated Python pipeline that tested and filtered 3,000+ model simulations using stability and out-of-sample validation, and designed a signal-combination method with built-in overfitting control

Research Data Scientist | National Taiwan University Taipei | Sep 2021 – Present

Ph.D. research, Interdisciplinary Neuroscience · Visiting Scholar, Brandeis University, USA (Apr 2024 – Jan 2025)

- Built end-to-end Python pipelines for two fMRI studies, including a 12-week randomised controlled trial, covering ~1.9 TB of 4D imaging data (483 scan sessions) from raw DICOM through preprocessing, parcellation, and whole-brain statistical inference
- Re-engineered the permutation-testing core with an algebraic reformulation of the regression model, cutting runtime from 2.1 days to 3.3 hours (~15×) with results verified identical to machine precision
- Led a 59-participant study end to end and coordinated a cross-site data platform between Taiwan and U.S. (Brandeis) teams, standardising data-quality checks and reporting for collaborators

Master's Researcher | National Taiwan University Taipei | Sep 2015 – Feb 2018

- Applied statistical modelling and structured data-acquisition workflows to multi-year clinical research datasets

SELECTED PROJECTS

- **Dissertation RAG Copilot** — Citation-grounded RAG system (hybrid retrieval, ChromaDB, hallucination verification) reaching 92% Hit@1 / 100% Hit@5; applicable to internal knowledge-base search (github.com/JamesLeeCY/paper-rag-copilot)
- **LINE Group Health Triage System** — Chinese-language NLP pipeline (response delay, sentiment, escalation signals) that ranks chat groups and outputs PDF reports so managers can prioritise client communications (github.com/JamesLeeCY/line-chat-triage)
- **Precision-Weighted Multi-Agent Investment Research System** — Taiwan-equity research system with five agents (fundamentals, news, macro/flows, supply chain, technicals), evaluated walk-forward on a survivorship-bias-free universe (271 stocks, 2019–2026, 8,475 predictions). Showed that precision weighting needs per-source accuracy estimates more stable than real data allows, and refocused the system on the fundamentals agent, whose within-sector stock selection held across sub-periods (IC +0.029, t = 2.16). Rules are frozen for pre-registered forward validation (github.com/JamesLeeCY/precision-weighted-investment-agent)

LEADERSHIP

- **Fundraising Co-Lead, BTBA Symposium 2025 — Boston Taiwanese Biotechnology Association (501(c)(3))** — Co-led a 6-person fundraising team for an annual two-day biotech symposium (350+ attendees); raised US$37K from 21 sponsors and donors, personally closing US$13K including the sole Diamond-tier sponsor; ran end-to-end sponsor operations (tiered packages, invoicing, payment tracking, receipts, benefit fulfilment)
- **Graduate Student President, National Taiwan University** — Represented the graduate cohort in academic and administrative coordination across departments

TECHNICAL SKILLS

Modelling: Regression and classification, time-series forecasting, hypothesis testing, factor design, walk-forward backtesting
Data & Reporting: Python (pandas, NumPy, scikit-learn), SQL, R, ETL pipelines, data-quality monitoring, dashboards and KPI reporting
AI: RAG, LLM agents, prompt engineering, PyTorch, Docker, GCP

EDUCATION

Ph.D., Interdisciplinary Neuroscience, National Taiwan University — 2021 – Expected 2027
M.S., Brain and Mind Sciences, National Taiwan University — 2015 – 2018
B.S., Forestry and Resource Conservation, National Taiwan University — 2011 – 2015
Awards: WorldQuant Summer Experience Program — Invited Participant, Certificate of Completion
Languages: Mandarin (native), English (professional working proficiency)